Smart Budget Planner

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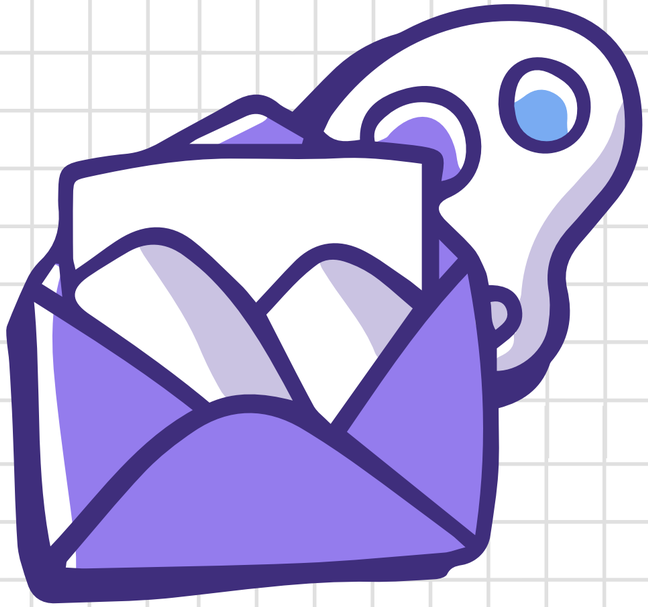
Problem Statement



Effective money management is a common struggle, often leading to overspending and missed savings goals. Without clear tracking, financial stress and wasted resources become inevitable. A streamlined budget planning tool empowers users to regain control by monitoring expenses and providing the data needed to make smarter, more intentional purchase decisions.

Project Objectives:

- Develop a Python-based budgeting system to record income and expenses.
- Automatically calculate and display the current balance after each transaction.

Reduce 2 objectives

Methodology

Graphical User Interface (GUI) Engineering

- Utilized the PySide6 framework to build a responsive desktop interface.
- Enhanced visual hierarchy and user experience through QSS (Qt Style Sheets).

Methodology

Data Persistence & JSON Management

- Developed a Data Access Layer using the JSON format to store transaction history locally.
- Created logic for CRUD operations (Create, Read, Update, Delete) to ensure that the user's income, expenses, and "Savings Vault" data are saved and retrieved accurately across sessions.

Methodology

Algorithmic Logic (Savings Vault)

- Implemented a Timer-based Algorithm using QTimer to simulate financial growth.
- Developed logic to ensure mathematical integrity, preventing balance deductions until the user-defined timer expires, thereby encouraging financial discipline.

Methodology

Testing & Debugging

- Conducted Unit Testing on individual functions (e.g., balance calculations and AI response parsing).
- Performed Integration Testing to verify that the GUI correctly reflects changes made to the underlying JSON database.

DEMO

Thank you